Phonetic-Aware Encoder Tuning: Boosting Robustness against L2 Pronunciation Variations via CTC Supervision

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Abstract

Recent advances in Automatic Speech Recognition (ASR) have achieved near-human performance for native speech through large-scale pre-trained models. Nevertheless, recognizing non-native (L2) speech remains a significant challenge, as "Phonetic Disruption" caused by L1 interference leads to substantial performance degradation. This study focuses on the Korean language to explore these challenges. We propose Phonetic-Aware Encoder Tuning, which applies LoRA (Low-Rank Adaptation) to the encoder of a Whisper Speech-To-Text (STT) model and incorporates an auxiliary Connectionist Temporal Classification (CTC) head to provide explicit acoustic-phonetic alignment.

We evaluate two phonetic representations—Korean Jamo (structural sub-character units) and IPA (international phonetic symbols)—as alternative auxiliary targets to enhance the encoder's acoustic resolution. Our experiments demonstrate that while both phonetic representations yield improvements, IPA-based supervision is particularly effective in reducing the Character Error Rate (CER) across different language groups. Detailed confusion matrix analysis confirms that our method robustly corrects language-specific error patterns, such as nasal coda confusion in Japanese and vowel height misidentification in Chinese and Vietnamese speakers.

Index Terms: speech recognition, acoustic and phonetic modeling, non-native speech processing

1. Introduction

Recent advancements in large-scale pre-trained models have enabled Automatic Speech Recognition (ASR) systems to achieve human-level performance for native speech [1, 2, 3]. However, for non-native (L2) speech, achieving robust recognition remains a significant challenge due to the phenomenon of L1 interference [4, 5, 6]. This occurs as L2 speakers often struggle to differentiate phonetic contrasts absent in their native language or project their existing articulatory habits onto the target language, resulting in a mismatch between actual acoustic signals and the standard distributions learned by the models. This study explores this acoustic-phonetic mapping failure—referred to as "Phonetic Disruption"—within the context of the Korean language.

A representative example—the misrecognition of an L2 speaker's "산업화 (San-eop-hwa)" (means industrialization) as "산오빠 (San-oppa)" due to articulatory weakening—illustrates how acoustic uncertainty leads to alignment errors. Since such recognition failures originate in the acoustic feature extraction stage prior to the decoder's linguistic inference, refining the encoder's representation is a fundamental and essential approach to addressing this issue. Therefore, we propose "Phonetic-Aware Encoder Tuning," a framework designed to preserve the pre-trained model's linguistic knowledge while precisely mapping distorted L2 acoustic patterns onto standard phonetic categories.

The proposed architecture applies LoRA (Low-Rank Adaptation) to the encoder blocks of a Whisper [2, 3] model and incorporates an auxiliary Connectionist Temporal Classification (CTC) head at the encoder output to perform multi-task learning. This framework processes the primary ASR task for final text generation in parallel with an auxiliary phonetic recognition task that predicts phonetically transcribed Korean sentences (g2pk2) [7, 8], Yale [9, 10], or IPA [11, 12] sequences, thereby encouraging the encoder to extract more discriminative acoustic features. While the auxiliary module guides the encoder's alignment during training, it is removed during inference, allowing the refined encoder representations to perform the final recognition without additional computational overhead.

To verify the effectiveness of the proposed method, we constructed an experimental dataset based on the NIKL Learner Corpus [13]. Leveraging the actual non-standard transcriptions of L2 speakers provided by the corpus, we implemented a data construction pipeline using XTTS [14] and the OpenAI API [15] to secure corresponding audio files and consistent ground-truth labels. This process yielded approximately 32,000 sentences across Japanese, Chinese, and Vietnamese L1 groups, featuring synthetic L2 speech that reflects specific articulatory habits alongside contextually accurate standard Korean labels.

Finally, this study demonstrates the quantitative improvements of the proposed framework through Character Error Rate (CER) and Jamo-level CER (Jamo-CER). Furthermore, we verify the acoustic-phonetic alignment via time-series probability distributions and utilize confusion matrix analysis to discuss how language-specific phonetic errors—such as nasal coda confusion in Japanese speakers or vowel height misidentification in Chinese and Vietnamese speakers—are robustly corrected at the acoustic representation level.

2. Proposed Method

In this section, we describe the proposed Phonetic-Aware Encoder Tuning framework, which enhances the Whisper [2, 3] architecture to mitigate phonetic disruption in L2 Korean speech. The overall architecture is illustrated in Figure 1.

2.1. Encoder-Focused Adaptation via LoRA

Since the primary characteristics of L2 speech errors—such as non-native accents and misarticulation—are primarily exhibited as acoustic deviations, we hypothesize that the feature extraction capability of the encoder must be refined rather than the linguistic reasoning of the decoder. To this end, we adopt an encoder-focused fine-tuning strategy while freezing the pa-

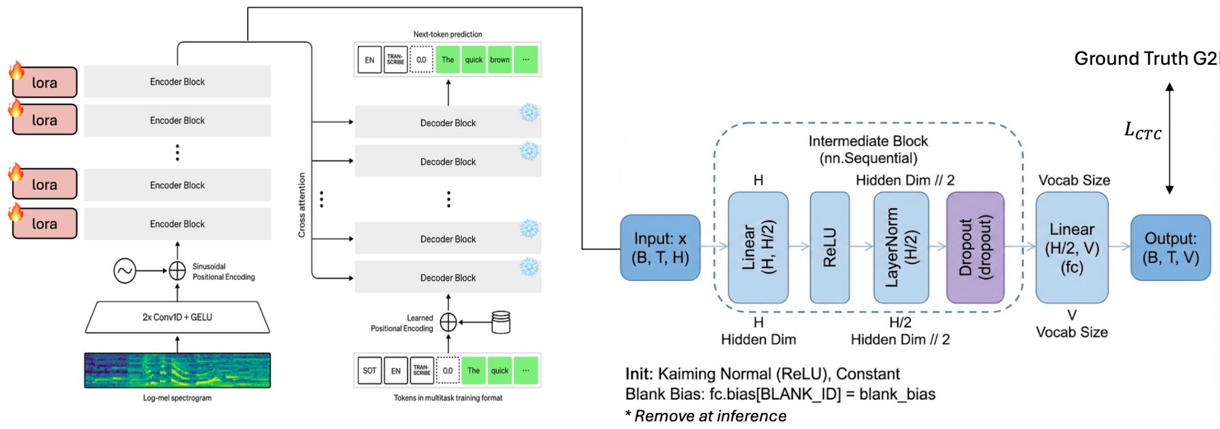


Figure 1: *Phonetic CTC LoRA framework.*

rameters of the decoder (Whisper-small). Freezing the decoder preserves the pre-existing linguistic knowledge of Korean and prevents catastrophic forgetting, which is crucial when training on limited L2 datasets. We apply Low-Rank Adaptation (LoRA) exclusively to the encoder blocks to efficiently capture L2-specific distorted acoustic patterns, such as improper tongue positioning and prosodic variations.

2.2. Multi-task Learning with Auxiliary CTC Head

To provide the encoder with explicit acoustic supervision, we incorporate an auxiliary phonetic recognition task into the training pipeline. We attach a lightweight Connectionist Temporal Classification (CTC) Head—consisting of a Linear-ReLU-Linear architecture—directly to the output of the encoder. This head predicts phoneme distributions for each acoustic frame, facilitating an automatic alignment between input speech and phoneme sequences.

The training follows a multi-task learning objective. Given an input sequence X and a target phoneme sequence Y, the CTC loss ($\mathcal{L}_{\text{CTC}}$) is defined by maximizing the log-likelihood of all possible alignment paths π that map to Y after the collapse operation $\mathcal{B}$:

$$P(Y \mid X) = \sum_{\pi \in \mathcal{B}^{-1}(Y)} P(\pi \mid X),$$

where $\mathcal{B}$ is the mapping function that removes blank tokens and merges identical consecutive phonemes. Since L2 speech often exhibits irregular phoneme durations, the monotonic alignment properties of CTC provide robust supervision for the encoder to map ambiguous signals onto standard phonetic categories. The final objective function is a weighted sum of the primary ASR loss and the auxiliary CTC loss:

$$\mathcal{L}_{\text{total}} = w_1 \mathcal{L}_{\text{ASR}} + w_2 \mathcal{L}_{\text{CTC}},$$

where w_1 and w_2 are hyperparameters that balance the importance of each task.

2.3. Phonetic Target: Phonetic Hangul, IPA, or Yale

To provide the encoder with explicit phonetic milestones, we propose a flexible framework that allows the selection of a target sequence from three distinct representations: Phonetic Hangul [7, 8], Yale [9, 10], or IPA [11, 12]. Instead of relying solely on standard grapheme-level supervision, the model learns to map physical acoustic signals to these phonetic or sub-character categories. This approach enables the encoder to develop higher acoustic resolution by choosing the most appropriate representation for the given research focus:

- **Phonetic Hangul (g2pk2)**: This target reflects the actual pronunciation of Korean by applying standard phonological rules. It is optimized for capturing errors in basic sound-to-letter mapping and structural combination errors across the onset, nucleus, and coda, which are frequent in L2 speech.
- **Yale Romanization**: Historically prominent in Korean linguistics, this target serves as a morphophonemic representation. It focuses on phonological structures with standardized Latin-based conventions, acting as an intermediate linguistic level that facilitates the learning of phonemic patterns.
- **IPA (International Phonetic Alphabet)**: This represents the most phonetically rigorous target, incorporating both phonological and detailed articulatory rules. By applying comprehensive phonetic transformations, it serves as the ground truth for analyzing fine-grained L2 pronunciation errors—such as aspiration and tensing—from a phonetician's perspective.

For instance, consider an L2 speaker's intention of "San-eop-hwa" (산업화), which might be misrecognized as "San-oppa" (산오빠) due to reduced articulation or liaison errors. While these segments may appear acoustically similar to a standard ASR model, their corresponding phonetic targets exhibit distinct differences:

- **Phonetic Hangul**: *사너퐈* [s a n eop hwa] vs. *사노빠* [s a n o ppa]
- **Yale**: `sanephwa` vs. `sanoppa`
- **IPA**: [sɑnʌp$^{\text{h}}$wa] vs. [sɑnop*ɑ]

By selecting the appropriate phonetic target and training via CTC loss, the encoder is guided to map these ambiguous acoustic patterns onto standard phonetic categories, leading to a more robust understanding of phoneme boundaries in L2 speech.

Table 1: *Performance comparison across L1 groups and phonetic targets. Values are reported as mean ± std. Lower is better. Underlined results indicate degraded performance compared to the original (baseline) model.*

Language	**Japanese (JP)**		**Chinese (CN)**		**Vietnamese (VN)**	
Model / Metric	**CER**	**Jamo-CER**	**CER**	**Jamo-CER**	**CER**	**Jamo-CER**
Original (Base)	0.1780 ± 0.2082	0.1171 ± 0.1661	0.1853 ± 0.2250	0.1247 ± 0.2517	0.1866 ± 0.3441	0.1253 ± 0.3538
LoRA + Jamo	0.1779 ± 0.2402	$\underline{0.1265 \pm 0.2032}$	0.1638 ± 0.1956	0.1048 ± 0.1445	0.1670 ± 0.3407	0.1149 ± 0.3502
LoRA + Yale	0.1514 ± 0.1994	0.0993 ± 0.1524	0.1759 ± 0.1982	0.1124 ± 0.1433	0.1641 ± 0.2252	0.1129 ± 0.2022
LoRA + IPA	0.1672 ± 0.4876	$\underline{0.1199 \pm 0.5550}$	0.1622 ± 0.1912	0.1044 ± 0.1350	0.1670 ± 0.3407	0.1149 ± 0.3502

3. Experimental Setup

3.1. Dataset and Augmentation Pipeline

To evaluate the proposed framework, we utilize the NIKL Learner Corpus [13], which contains diverse non-standard transcriptions of Korean L2 speech. We focus on the three most frequent L1 groups—Japanese (JP), Chinese (CN), and Vietnamese (VN)—to ensure statistical robustness, selecting over 10,000 sentences for each language.

A critical challenge of the NIKL corpus is the absence of corresponding audio and consistent sentence-level ground truth (GT). Existing corrections are often fragmented; for instance, while a phonetic error like *"san-oppa"* is corrected to its standard form *"san-eop-hwa"*, other contextual errors in the same sentence frequently remain unaddressed. To bridge this gap, we implemented a dual-stage augmentation pipeline:

1. **L2 Speech Synthesis**: We leverage the XTTS model [14] to synthesize speech from the non-standard L2 transcriptions. By utilizing native speaker audio as a speaker prompt, the model generates synthetic speech that reflects the characteristic articulatory habits and L1 interference patterns of each language group.
2. **Ground-Truth Normalization**: We employ the OpneAI API [15] to reconstruct fragmented labels into coherent, sentence-level ground-truth. By providing context-aware prompts, we normalize non-standard L2 strings into standard Korean, ensuring high-fidelity labels even for words missing explicit correction in the original corpus.

The resulting dataset comprises 10,672 JP, 12,278 CN, and 9,090 VN samples, which are partitioned into training, validation, and test sets using an 8:1:1 ratio.

3.2. Implementation Details

We trained nine independent models to cover all combinations of the three L1 groups and three phonetic targets (Phonetic Hangul, Yale, and IPA). We adopted Whisper-small [2] as our backbone architecture. Following our proposed multi-task learning objective, the weights for the combined loss function were empirically set to $w_1 = 0.7$ (ASR) and $w_2 = 0.3$ (CTC).

Training was conducted for 10 epochs using a batch size of 4 and a fixed learning rate of 1×10^{-4}. To isolate the impact of our phonetic-aware tuning, all other hyperparameters were kept constant across all language-specific experiments.

3.3. Evaluation Metrics

The performance of our models is assessed using two metrics: Character Error Rate (CER) and **Jamo-level CER (Jamo-CER)**. While CER provides a general measure of transcription accuracy, it is often insufficient for Korean L2 speech analysis because it treats each syllabic block as an atomic unit, thereby failing to capture sub-syllabic phonetic misalignments.

Given that Korean Hangul is a featural alphabet grouped into syllabic blocks, Jamo-CER is essential to reflect the phonetic distance between prediction and ground truth. This metric decomposes each character into its constituent sub-character units: onset, nucleus, and coda. For instance, a substitution error between "gan (간)" and "gang (강)" is treated as a total character failure in standard CER, whereas Jamo-CER correctly identifies it as a partial (1/3) phonetic mismatch in the coda. This allows for a more granular analysis of articulatory failures—such as the coda nasal confusion common in Japanese speakers—that would otherwise be obscured at the character level.

4. Results

4.1. Quantitative Analysis

The experimental results across three L1 groups (Japanese, Chinese, and Vietnamese) demonstrate that our phonetic-aware encoder tuning significantly improves ASR performance on L2 Korean speech. Table 1 summarizes the Character Error Rate (CER) and Jamo-level CER (Jamo-CER) for each configuration.

As shown in Table 1, the proposed phonetic-aware tuning generally led to performance gains across the majority of experimental configurations. Both CER and Jamo-CER were improved in most cases, particularly for the Chinese (CN) and Vietnamese (VN) groups. However, a notable exception was observed in the Japanese (JP) group, where certain phonetic targets resulted in degraded performance. Specifically, the Jamo-CER for the JP group increased to 0.1265 (LoRA+Jamo) and 0.1199 (LoRA+IPA) compared to the baseline's 0.1171, as indicated by the underlined values. This suggests that while auxiliary phonetic supervision typically enhances acoustic resolution, its effectiveness can vary depending on the specific phonological characteristics or alignment challenges inherent to a particular L1-L2 pair.

4.2. Qualitative Analysis

To further analyze the impact of phonetic-aware tuning, we compared the inference outputs of the baseline model and proposed IPA-targeted model. Table 2 presents selected examples demonstrating the correction of common L2 phonetic errors.

The qualitative results in Table 2 further elucidate the behavior of the proposed IPA-targeted model. The upper section illustrates successful recovery cases where the model correctly predicts the standard form despite significant acoustic distortion in L2 speech. For instance, the model successfully distinguishes the semantic target *San-eop-hwa"* (산업화) from the acoustically similar *San-oppa"* (산오빠) and corrects lenition errors (e.g., *Tae-eo-na"* vs. *Dae-eo-na"*), demonstrating that the auxiliary phonetic supervision effectively sharpens the encoder's acoustic resolution.

Table 2: *Inference examples from L2 Korean speech. ForestGreen indicates Ground Truth (GT), Red indicates Baseline errors, and Blue indicates Proposed model corrections.*

Transcription (GT / Baseline / Proposed)
Successful Correction Examples
그렇게 해서 산업화가 됐다고 들었습니다 / 그렇게 해서 산오빠가 됐다고 들었습니다 / 그렇게 해서 산업화가 됐다고 들었습니다
중국의 하얼빈에서 태어나셨어요 / 중국의 하얼빈에서 대어 나셨어요 / 중국의 하얼빈에서 태어나셨어요
그 자살이 나 왕따 그런 문제도 / 그 자살이 나완다 그런 문제도 / 그 자살이 왕따 그런 문제도
지금 필요 없다고 생각합니다 / 지금 비료 없다고 생각합니다 / 지금 필요 없다고 생각합니다
네 그 상사도 나랑 안 맞는 사람이라서 / 네 그 산사도 나랑 안 맞는 사람이라서 / 네 그 상사도 나랑 안 맞는 사람이라서
Challenging or Modified Examples
수족 냉증이 심해서 / 수족 렌치니 시임에서요 / 수족 냉층이 시임에서
남한은 대통령이 계시고 / 남한은 대통령이 계시고 / 나마는 대통령이 계시고

However, as shown in the lower section, some cases exhibit degraded or modified outputs. We speculate that these instances, as well as the sub-optimal performance (underlined) observed in Table 1, are likely attributable to the inherent noise in the NIKL corpus. Despite our normalization efforts, certain fragmented or inconsistent ground-truth labels in the original dataset may have misguided the phonetic alignment during the multi-task learning process. These erroneous labels can force the model to over-adapt to non-standard transcriptions, resulting in phonetic outputs that deviate from standard Korean orthography even when the acoustic signal is relatively clear.

5. Analysis

5.1. L1-Specific Phonetic Improvement Patterns

To investigate the impact of phonetic-aware tuning on language-specific interference, we analyzed the changes in the Top 20 most frequent phonetic errors for each L1 group using error-change matrices. For each language, the Top 3 most improved phonetic pairs were identified, reflecting the distinct phonological characteristics and interference patterns of each native language.

- **Japanese (JP)**: Japanese speakers often struggle with Korean coda nasals and the boundary between plosives and fricatives due to their native syllable structure. Key improvements include:
  - **[NG] → [N] (-26)**: The ambiguous articulation of the Japanese nasal 'n' is more accurately aligned by the encoder to the Korean velar nasal (ㅇ).
  - **[H] → [NG] (-22)**: Correction of nasalization errors where the friction signal of 'h' was misprocessed as a nasal.
  - **[NG] → [M] (-16)**: Improved distinction between labial-closing [M] and velar [NG] in the coda for speakers accustomed to open syllables.
- **Chinese (CN)**: Chinese learners frequently face challenges with the Korean [u]/[o] vowel system and labial codas. Significant improvements were observed in:
  - **[u] → [o] (-29)**: Resolving misidentification of the Chinese-style 'u' (lacking lip rounding) by classifying it correctly into the standard Korean [u] category.
  - **[M] → [NG] (-19)**: Restoring labial nasals [M] in the coda, which were previously substituted with the more familiar [NG].
  - **[eo] → [o] (-17)**: Enhanced differentiation of tongue height for the Korean mid-vowel [eo].
- **Vietnamese (VN)**: Vietnamese speakers exhibit weaknesses in tongue height differentiation and syllable-initial fricative strength. Notable gains include:
  - **[eo] → [o] (-37)**: Resolving the ambiguous tongue height boundary caused by the complex Vietnamese vowel system.
  - **[N] → [NG] (-26)**: Robust alignment of unclear coda nasal articulation to standard targets.
  - **[H] → [NG] (-22)**: Guarding against the tendency to nasalize or omit the syllable-initial fricative [H].

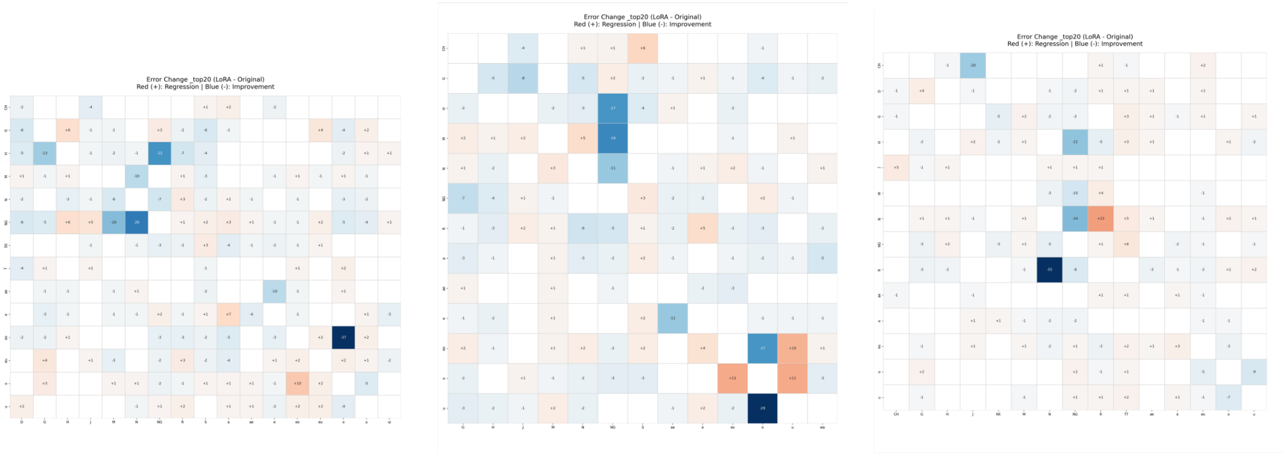
5.2. Timestamp-based Peak Alignment Analysis

To verify whether the auxiliary CTC head facilitates precise phonetic timing, we conducted a forced alignment analysis using timestamped peaks. As illustrated in Figure 3, the original model's probability for correct phonetic units is diffused across multiple frames, indicating substantial confusion during interpretation. In contrast, the proposed framework allows the encoder to establish sharp probability peaks precisely at the ground-truth positions, despite the presence of acoustic imperfections in L2 speech. This demonstrates that the auxiliary phonetic supervision effectively forces the encoder to map ambiguous signals onto standard phonetic categories. Such a shift indicates that the acoustic interpretation has been fundamentally refined at the encoder level through the multi-task learning task.

6. Conclusion

This study proposed a Phonetic-Aware Encoder Tuning framework that integrates LoRA-based adaptation with an auxiliary CTC head to mitigate phonetic disruptions in L2 Korean speech recognition. By implementing a multi-task learning task with explicit phonetic targets—most notably IPA, which proved superior in providing high-resolution articulatory milestones—the model successfully guided the encoder to extract more discriminative acoustic features across Japanese, Chinese, and Vietnamese speaker groups.

However, the absence of high-fidelity ground truth (GT) and the fragmented nature of corrections in the NIKL Learner Corpus emerged as a primary limitation. The observed performance degradation in specific metrics, such as the Jamo-CER for the Japanese group, suggests that the auxiliary module occasionally over-adapted to inconsistent or noisy phonetic labels inherent in the original dataset. These findings highlight that while encoder-level phonetic supervision is a powerful strategy for resolving L1 interference, its ultimate success remains heavily dependent on the availability of human-validated, high-quality L2 data to ensure accurate phonetic alignment.



Japanese (JP) *Chinese (CN)* *Vietnamese (VN)*

Figure 2: *Phonetic error change matrices (LoRA – Original) for the top-20 most frequent phonetic errors across L1 groups. Blue cells indicate a reduction in specific phonetic errors compared to the baseline.*

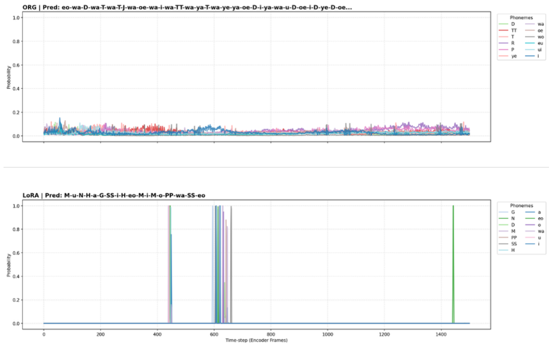


Figure 3: *Forced alignment analysis showing the probability distribution of phonetic units. The original model (top) exhibits diffused peaks, while the proposed model (bottom) generates sharp, correctly aligned peaks at the ground-truth timestamps.*

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